

Memcached (High-performance distributed memory caching system)

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What is Memcached?

- Memcached is an **in-memory key-value** caching system
- Used to speed up **dynamic web applications** by reducing database load.

Key Features

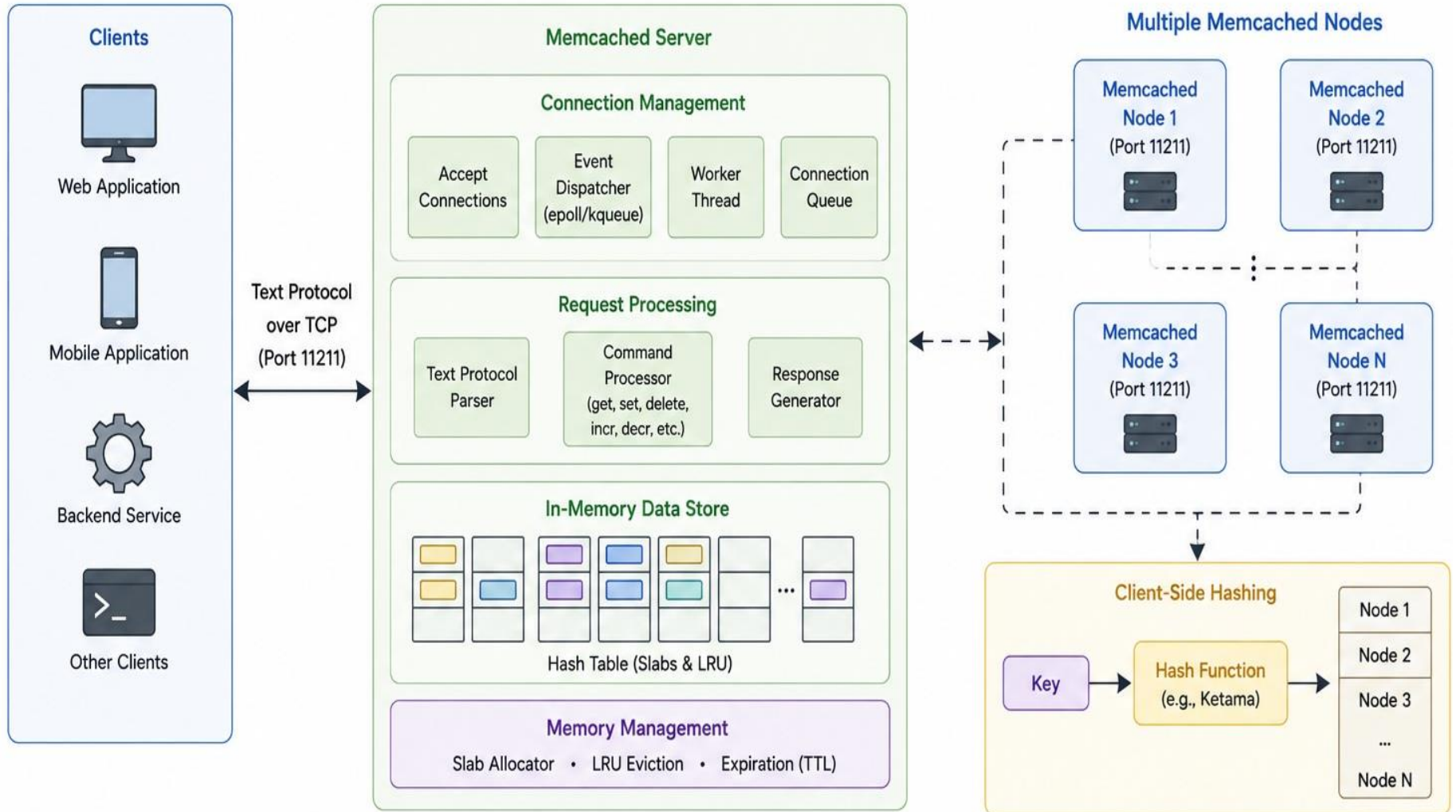
- - In-memory storage for fast access
- - Distributed architecture
- - Simple key-value data model
- - No persistence
- - High scalability

Architecture

- Client → Memcached Server Pool → Database
- Clients hash keys to determine server location.

Architecture

Memcached Architecture



How Memcached Works

1. Client requests data
2. Cache checked first
3. If miss → fetch from DB
4. Store result in cache
5. Serve future requests from cache

Data Model

- - Key: String
- - Value: Ramesh (any data)
- - Expiry time supported
- - No complex queries

Use Cases

- Web session caching
- Database query caching
- API response caching
- Page rendering acceleration

Limitations

- - No persistence
- - No replication by default
- - Data loss on restart
- - Limited data structures

Example (Python)

```
import memcache  
mc = memcache.Client(['127.0.0.1:11211'])  
mc.set('key', 'value')  
print(mc.get('key'))
```

Memcached vs Database

- - Memcached: Fast, temporary storage
- - DB: Persistent, slower
- - Used together for performance optimization

Advantages

- - Extremely fast (RAM-based)
- - Simple design
- - Horizontal scaling
- - Reduces DB load

- Thank You